US Treasury Yield Curve: Principal Component Analysis

Methodology

We collected daily US Treasury constant-maturity yields from the FRED database for ten maturities: 3 months, 6 months, 1 year, 2 years, 3 years, 5 years, 7 years, 10 years, 20 years, and 30 years. The sample covers January 2010 to October 21, 2026, providing more than 16 years of historical observations. We applied PCA to the daily yield changes using Python's scikit-learn library.

For the Principal Component Analysis, daily yield changes were calculated as: $\Delta y_{t,m} = y_{t,m} - y_{t-1,m}$

PCA Results

The first three principal components explain 95.74% of the total variance in daily US Treasury yield changes.

Principal Component	Explained Variance	Cumulative Variance
PC1 — Level	80.11%	80.11%
PC2 — Slope	11.70%	91.81%
PC3 — Curvature	3.94%	95.74%

Figure 1. Explained Variance of the First Three Principal Components

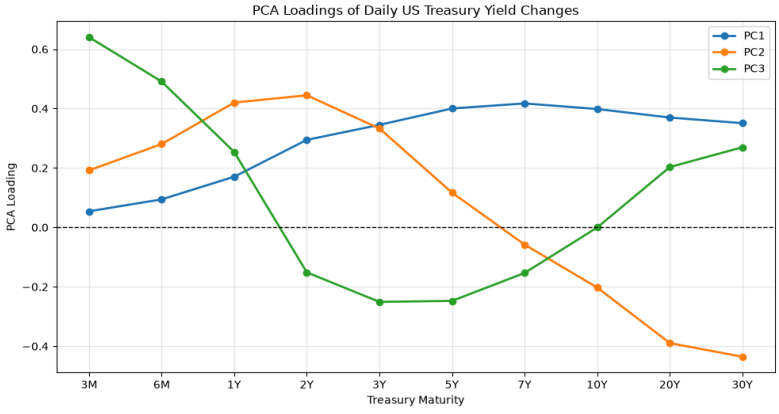
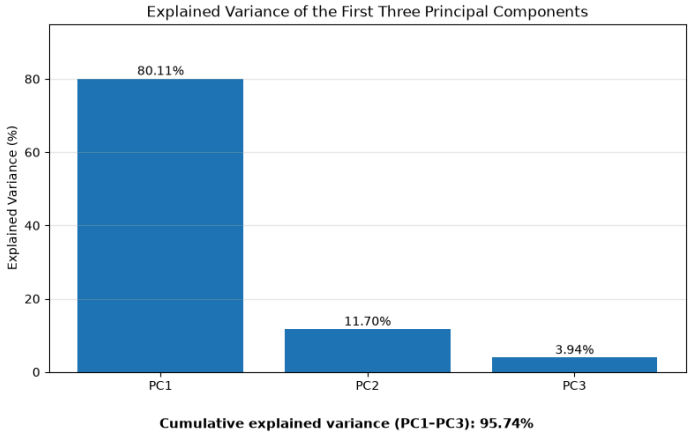


Figure 2. PCA Loadings of Daily US Treasury Yield Changes Three Principal Components



The loadings provide an economic interpretation of the three components:

PC1 - Level (80.11%): All maturities have positive loadings, showing that yields move in the same direction. However, the loadings are larger for medium- and long-term maturities, meaning that the movement is not a parallel shift.

PC2 - Slope (11.70%): Short-term maturities have positive loadings, while long-term maturities have negative loadings.

PC3 - Curvature (3.94%): Short and long-term maturities have positive loadings, whereas intermediate maturities have negative loadings. The middle of the curve moves differently from its shorter and longer ends.

Discussion and Conclusion

Our results support the three-factor interpretation of Treasury yield curve movements discussed in Section 6 of the lecture. The first three components jointly explain 95.74% of the variation, matching the approximately 95% benchmark in the course material, which means PCA provides a useful framework for understanding and monitoring the structure of interest-rate risk.

References

- Federal Reserve Bank of St. Louis, FRED — Federal Reserve Economic Data. Daily US Treasury constant-maturity yield series: DGS3MO, DGS6MO, DGS1, DGS2, DGS3, DGS5, DGS7, DGS10, DGS20, and DGS30. https://fred.stlouisfed.org/